



Physical domain knowledge based image processing algorithm for in-situ surface porosity inspection and digitalization for 3D concrete printing structures

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ABSTRACT

Concrete 3D printing holds transformative potential for construction because it can enable rapid, design-flexible structures from terrestrial housing to extraterrestrial habitats. However, its scalability is limited by challenges in quality control which governs structural integrity and durability, among which the volume fraction and size distribution of macro-porosity are especially critical. This study presents a novel scientific in-process monitoring technique for real-time macro-porosity analysis in 3D-printed concrete extrudates. Using a microscopy system with a side-mounted light source, the monitoring system captures surface microstructure without labor-intensive petrographic post-processing. A self-calibrating computer vision algorithm, leveraging a Perimeter-to-Area Ratio (PAR) metric and concrete domain knowledge, accurately distinguishes air voids from aggregates, overcoming conventional image processing misclassification. Validated against expert annotations, the algorithm achieved a 98% porosity prediction accuracy relative to manual petrographic baselines. Furthermore, the bimodal thresholding successfully mathematically isolates small-entrained spherical pores (<1.0 mm) from large-entrained irregular voids (>1.25 mm) without manual input. This precise pore geometry and distribution analysis provides the actionable data required to optimize mixture designs on-the-fly, forecast a material's resistance to environmental stressors like freeze-thaw cycles, and enable the creation of digital twins for predictive maintenance. While challenges remain in adapting to diverse concrete compositions and variable lighting, this method provides a scalable, quantitative foundation for reliable 3D printing, advancing additive manufacturing toward sustainable, industry-standard construction.

Acronyms & abbreviations

ASTM	American Society for Testing and Materials
CNN	Convolutional neural network
PAR	Perimeter to area ratio
PAR _e	PAR of small-entrained pores
PAR _E	PAR of large-entrained pores
PORECAST	Porosity Forecast System
SEM	Scanning Electron Microscopy
TRL	Technology Readiness Level
UHPC	Ultra-High-Performance Concrete

1. Introduction

Concrete 3D printing has the potential to revolutionize construction by enabling rapid, cost-effective, and design-flexible structures for applications ranging from affordable housing to lunar habitats. However, its manufacturing readiness is currently limited by the complex rheological properties of concrete mixtures—generically characterized by their flowability, extrudability, and buildability. These properties vary significantly with environmental factors, such as temperature and humidity, as well as component material properties and their volumetric proportions [1–9]. Crucially, these factors shape the printed material's microstructure, particularly its porosity, which dictates both structural integrity and durability. Controlled macro-porosity (pores ≥ 1 mm in

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characteristic dimension) filled with air can mitigate microcrack formation during freeze-thaw cycles by facilitating pressure release (Fig. 1a) [10–12]. In 3D-printed concrete, this intentionally introduced “air void system” enhances durability. Conversely, unintended pores or cracks of similar or larger sizes can arise from poor mixing, inhomogeneous extrusion, plastic shrinkage, or chemical reactions [7,8]. These defects compromise structural integrity and increase material waste, underscoring the critical need for precise, real-time porosity control and monitoring.

Traditional porosity assessment methods are ill-suited for the demands of 3D printing. Optical petrography, which involves slicing, coloring, and scanning samples for one-dimensional chord length measurements (Fig. 1b), is laborious and unsuitable for real-time monitoring or ex-situ inspection [12–14]. While advanced characterization techniques such as Scanning Electron Microscopy (SEM) [15–17], X-ray imaging [3,18–22], and micro-CT [4,23] provide highly detailed void analyses, their complexity and cost render them impractical for in-process or offline field deployment.

To bridge this gap, recent advances in optical sensing and computer vision offer promising avenues for smart manufacturing in 3D-printed concrete [24]. For instance, Silvia et al. [25] trained machine learning models to predict extrudate width based on printing speed and geometry, while Garcia et al. [26] utilized vision systems for real-time defect identification by automatically detecting variations in layer thickness and texture. Furthermore, Beskopylny et al. [27] employed a U-Net model with cellular automata to detect macro-scale microstructural defects (> 3 mm), and Mirbod and Shoar [28] applied pattern recognition for large crack detection. Despite these strides, the accurate characterization of smaller, performance-critical pores (< 1 mm) remains underexplored.

A major hurdle in evaluating these smaller pores is the inadequacy of conventional image processing, which frequently misclassifies dark aggregates as air voids. This issue is inherent to basic threshold-based algorithms applied to uncolored micrographs and is distinct from the manual ASTM C457 procedure, which avoids misclassification by coloring the voids prior to measurement. This misclassification artificially inflates porosity estimates—for example, erroneously registering 54.9% porosity in a light paste sample that actually contains only 5.6% porosity (Fig. 2a1–a3)—or fails entirely to differentiate voids in dark paste with light aggregates (Fig. 2b1–b2); the concrete microstructure images were sourced from [29] processed with off the shelf contour finding computer vision algorithm (based on the Suzuki method [30]). To address these fundamental measurement barriers, this study introduces PORECAST.

Developed herein, PORECAST represents a novel, scientific approach to real-time porosity monitoring that combines specialized hardware

with domain-specific algorithms, emphasizing a research-driven design tailored to overcome the limitations of both traditional petrography and conventional vision frameworks.

2. Methods

This system integrates a novel hardware-algorithm pipeline to monitor extrudate porosity in real-time 3D-printed concrete. It combines optical microscopy with a self-calibrating computer vision pipeline to overcome limitations of conventional petrographic image analysis. This section describes the experimental setup, computer vision pipeline, threshold optimization with concrete domain knowledge, and expert validation.

2.1. Experimental setup

The system was tested using a cured cast UHPC cube without fibers, simulating extrudate surfaces without additional processing. A microscope with a side-mounted light source captured microstructure images (Fig. 3a). The light source, positioned at a 2-degree angle to the specimen surface, was used to standardize the lighting environment with a consistent white balance across day or night printing and cast shadows to distinguish deep air voids from flat aggregates (Fig. 3b), eliminating the need for coloring resins.

2.1.1. Computer vision pipeline

The pipeline quantifies pore size, shape, and distribution while avoiding aggregate misclassification. Images undergo grayscale-to-binary conversion using a threshold, followed by contour detection via the Suzuki method [30], the latter being selected for its robust closed-contour output over Canny's algorithm, which requires two hyperparameters and produces open contours [34].

Porosity estimates are sensitive to threshold selection, causing fluctuations in pore area and count (Fig. 4). Most computer vision methods for porosity analysis rely on well white-balanced images and use either mid-point grayscale thresholds, manually tuned thresholds based on petrographic expertise, or calibration with X-ray-derived porosity ground truth [22,35,36]. To overcome these limitations, our system's algorithm optimizes thresholds using a Perimeter-to-Area Ratio (PAR) metric, defined as:

$$PAR = \frac{P}{\sqrt{A}} \quad (1)$$

where P is the perimeter of the contour and A its area. The PAR reaches a

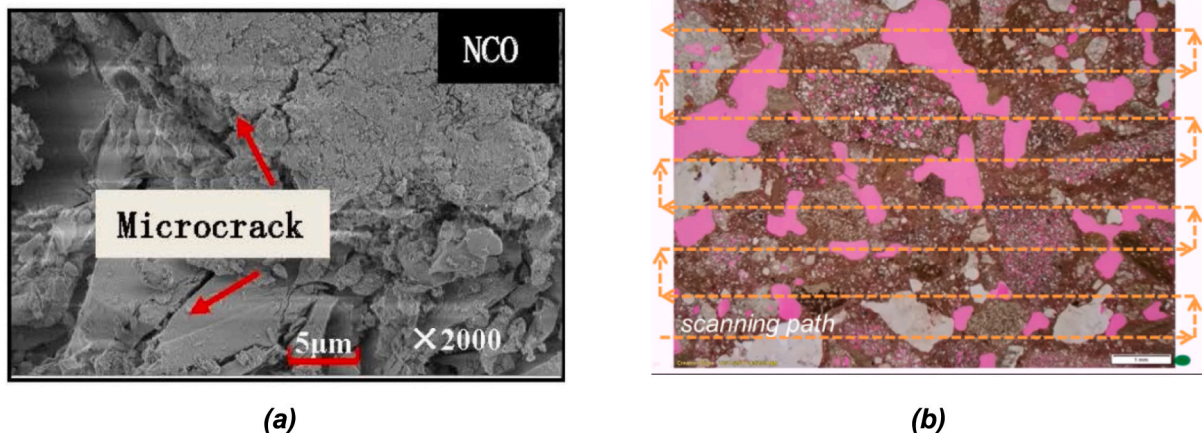


Fig. 1. (a) Microcracks after 30 freeze-thaw cycles due to insufficient protective pores (adapted from [31]). (b) Chord length measurement of the colored air voids with a rectilinear scanning path (orange dotted lines) as described in the ASTM C457/C457M-16 standard [32] (processed microscope image adapted from online petrography lecture [33]).

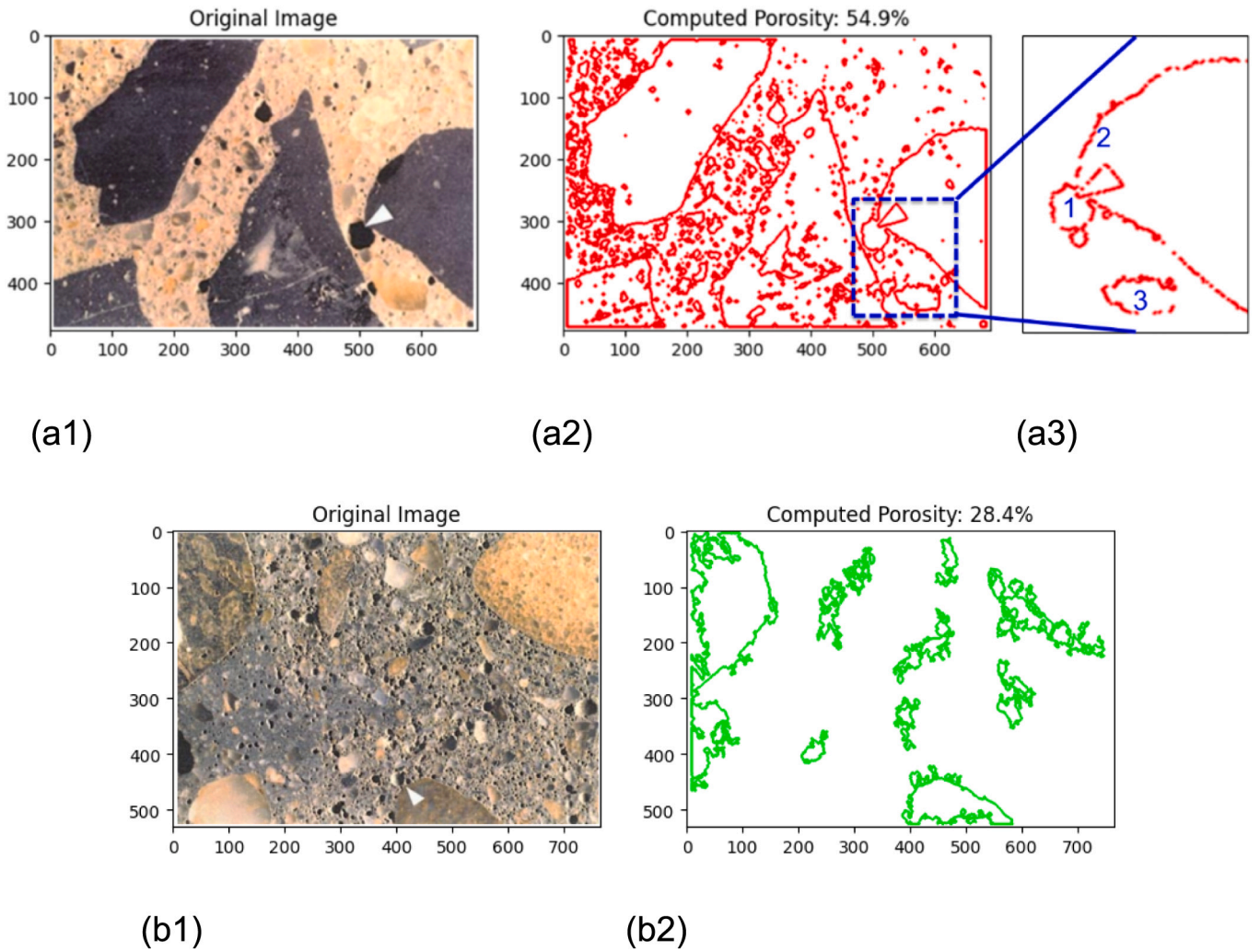


Fig. 2. a1) Not post-processed microstructure surface of finely lapped slice of concrete containing 5.6% porosity that when a2) image processed was overestimated at 54.9% and could not achieve a3) differentiation of air-voids, dark aggregates and sand particles. b1) A similar instance of a surface with darker paste, light-colored aggregates and small dark air-voids b2) which also, when processed, overestimated the present porosity. Note: x- and y-axes in the microstructure images are pixel counts.

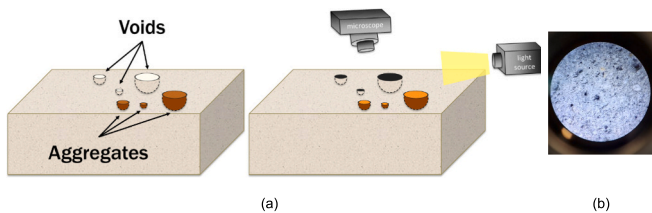


Fig. 3. (a) Illustration of the monitoring system, showing how the microscope captures the void images due to their depth when paired with a light source that is illuminating the surface from the side (b) the microstructure capture of an UHPC sample.

minimum of $2\sqrt{\pi}$ for circles and increases without bound as the aspect ratio increases.

2.2. Threshold optimization and pore classification

Our method for threshold optimization uses concrete domain knowledge, categorizing pores into capillary voids ($\leq 5 \mu\text{m}$, undetectable optically), entrained air voids ($< 1 \text{ mm}$, spherical), and entrapped/water voids ($\geq 1 \text{ mm}$, irregular) [37]. The pipeline calibrates thresholds to

maximize detection of spherical entrained pores, using their binomial distribution as the primary indicator (Fig. 5). A double Gaussian function fits the pore count data to identify the optimal threshold:

$$f(x) = a_0 + a_1 e^{-\frac{(x+b_1)^2}{2c_1^2}} + a_2 e^{-\frac{(x+b_2)^2}{2c_2^2}} \quad (2)$$

where the constants a_0, a_i, b_i, c_i determine the shape of the double Gaussian distribution. The first Gaussian peak identifies a binarization threshold where only entrained pores ($< 1 \text{ mm}$, spherical) are detected in the processed image. As the threshold increases, partial imaging of entrapped pores ($> 1 \text{ mm}$, irregular) causes their shapes to be misclassified as circular, creating a mixed regime. To avoid this, the threshold at the second Gaussian peak is used, ensuring a “pure” region where entrained pores are accurately depicted as circular (Fig. 6b). This calibration maximizes detection of spherical pores while minimizing misclassification of irregular entrapped pores. Pixel limits ($16 \mu\text{m}$ for small-entrained pores, ~ 10 pixels; 1.25 mm for large-entrapped pores) ensure accurate categorization (Fig. 7). Four rules refine threshold selection:

1. Select regions with continuous, stable PAR values, indicating focused pore geometry.

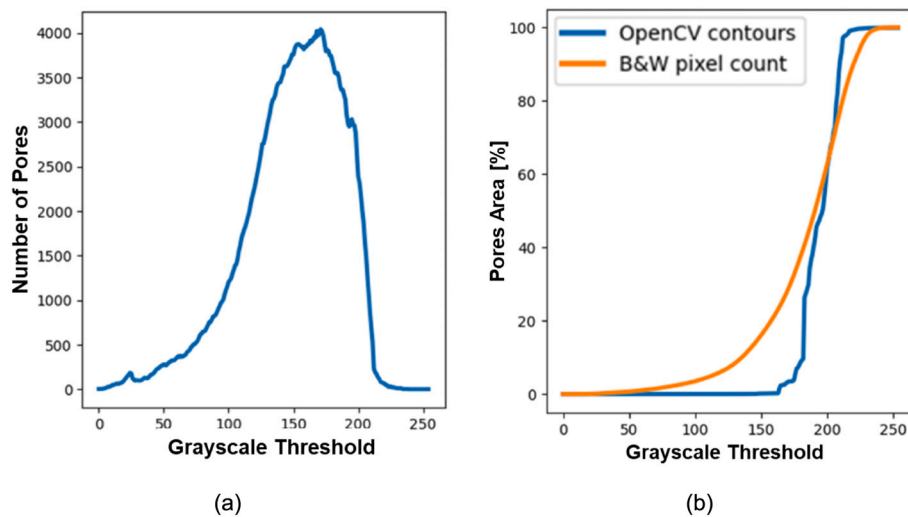


Fig. 4. The threshold value used to produce the binary image affects the subsequent computed porosity properties: graph (a) shows the number of identified pores and (b) the computed percentage pore area, with a simple black and white pixel count (orange) and with the Suzuki method (blue) versus the thresholding value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

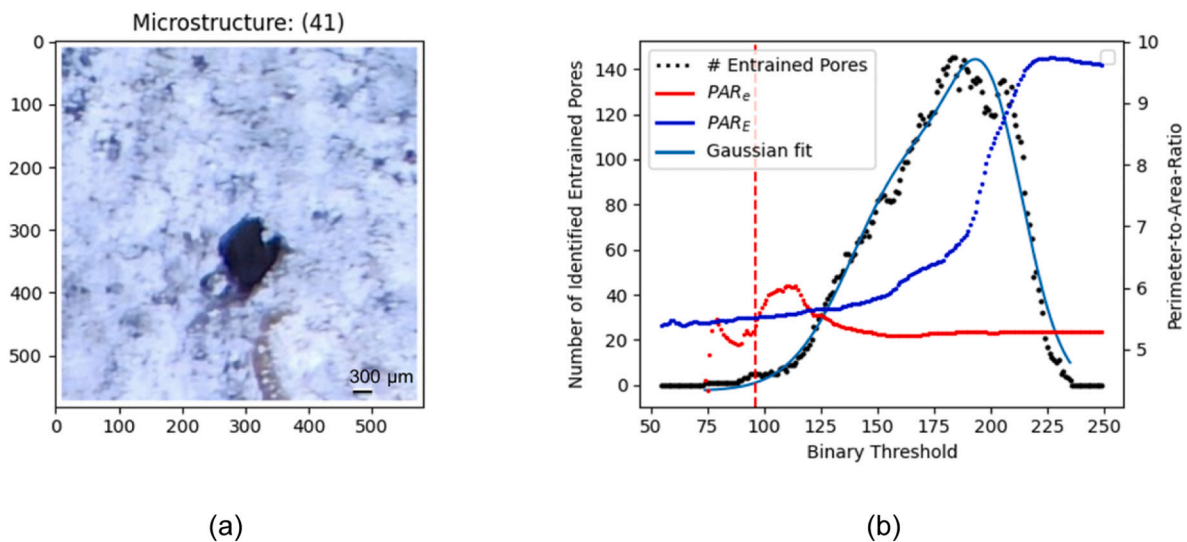


Fig. 5. (a) The microstructure image of section (4,1) and (b) the characteristics of the entrained and entrapped pores versus the binary imaging threshold. Three main computed properties are used to characterize the pores: the average PAR ratio of the small-entrained and of the large-entrapped pores and the total number of small-entrained pores.

2. Include areas where $PAR_E > PAR_e$, reflecting irregular entrapped pores.
3. Exclude high-threshold regions (~ 150) to avoid mixed pore misclassification.
4. Maximize entrained pore count for optimal spherical void detection (Fig. 8).

Optimizing the binary threshold this way can be challenging for some microstructure images, which then complicates the accurate detection of small-entrained pores (< 1 mm). Fig. 9 shows three images where the fitting process struggles. In Fig. 9a, the Perimeter-to-Area Ratio (PAR) for large-entrapped pores (> 1.25 mm) is discontinuous at thresholds < 125 , which means that pore geometry detection is unstable. For that image Rule 1 (Section 2.3) is used to prioritize continuous PAR regions. Fig. 9b shows noisy data at thresholds between 140 and 180, which leads to misplaced peaks in the Gaussian model. Rule 3 is used to exclude high-threshold mixed zones in these scenarios. In Fig. 9c, the leftmost peak, which is critical for detection of small-entrained pores, is

nearly obscured by the dominant second peak. This overlap is driven by high signal-to-noise ratios and requires manual correction using visual inspection and the four rules outlined in Section 2.3. All three cases in Fig. 9 underscore the need for domain knowledge integration and highlight future improvements, such as advanced fitting algorithms, to enhance the system's robustness for real-time applications.

2.3. Methodological framework

To clarify the experimental design and the flow of data through the PORECAST system, Table 1 outlines the overarching methodological framework of this study. It details the variable parameters utilized during the physical setup and image acquisition, the dynamic variables processed by the self-calibrating algorithm, and the final microstructural indicators determined by the system.

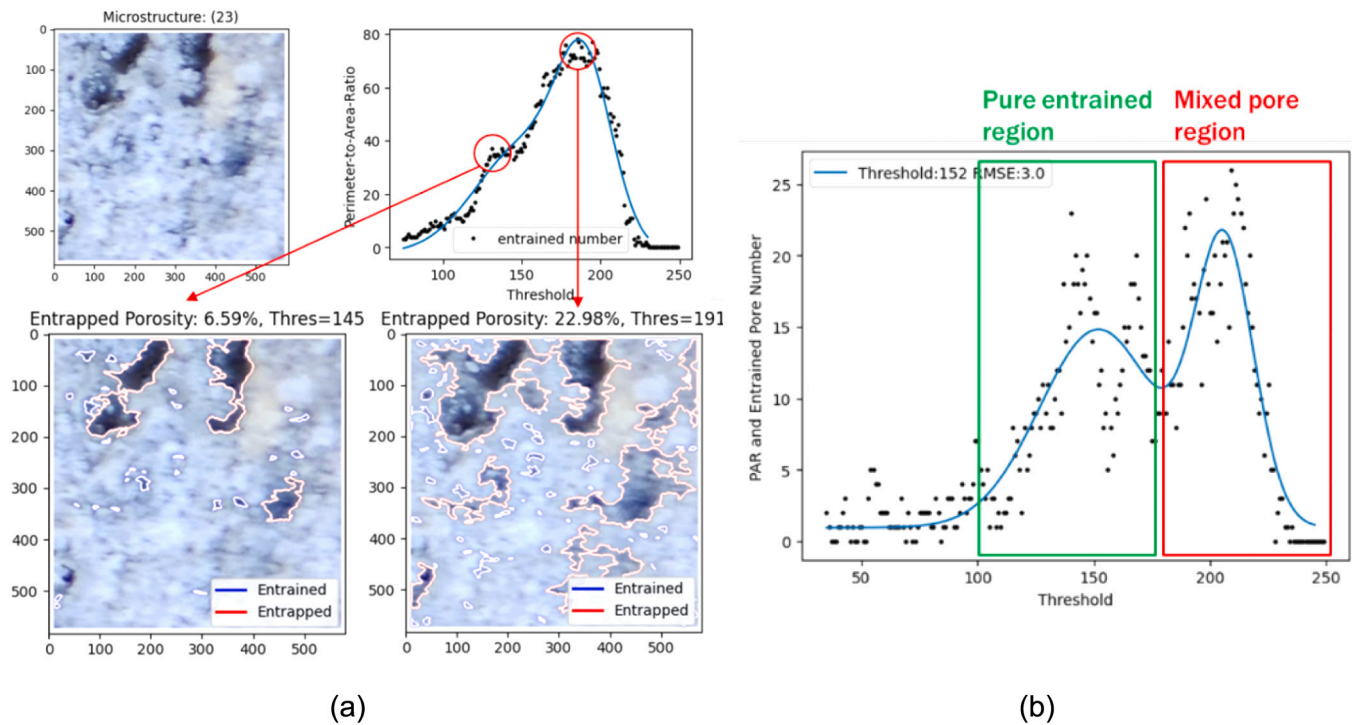


Fig. 6. (a) Exemplary binomial distribution fits the number of small-entrained pores below 1 mm at the two peaks of the individual normal distributions Note: x- and y-axes in the microstructure images are pixel counts (b) and the two regions corresponding to the pure entrained region and the mixed entrained-entrapped region.

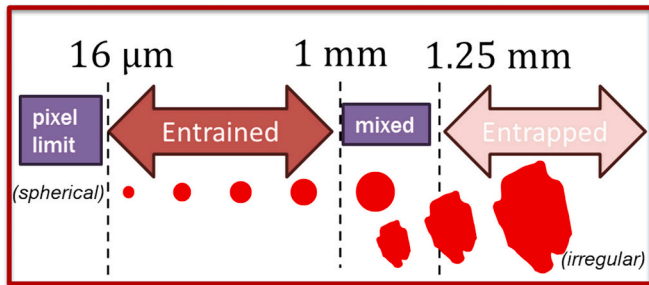


Fig. 7. Pore type bounds used in the algorithm to classify based on their size (diameter).

2.4. Validation with expert knowledge

The algorithm was validated by comparing porosity estimates to expert annotations on the UHPC image of 29×29 mm divided into a 5×5 grid (Fig. 10a). Petrography experts outlined large-entrapped pores (>1.25 mm) in red (RGB: 255, 0, 0) to test boundary delineation in challenging cases, such as shared aggregate-pore boundaries (Fig. 9b) or overlapping voids (Fig. 10c). Results are detailed in Section 3.

3. Results

This section presents the algorithm's accuracy, threshold variability, pore boundary delineation, and performance in challenging cases, demonstrating its potential to enable real-time porosity monitoring in 3D-printed concrete extrudates of UHPC. The average algorithm porosity prediction of all pores among all microstructure images exhibits a 2% error. The algorithm correctly identified the number and location of the largest entrapped pores (>1.25 mm) for most microstructure splits.

3.1. Threshold variability

The optimal binary threshold, derived through a double Gaussian model (Eq. (2)), has significant variations across microstructure images due to differences in concrete paste composition, aggregate distribution, and image intensity. As shown in Fig. 13 (a), a histogram of the algorithm's optimal thresholds (8-bit grayscale values of 0 to 255) has a mean of 129 and a standard deviation of 27.9. This variability emphasizes the need for a self-calibrating step to dynamically adapt to extrudate heterogeneity. This adaptability ensures robust porosity characterization in real-time 3D printing applications, eliminating reliance on fixed thresholds and enhancing reliability across diverse concrete mixtures and environmental conditions.

3.2. Pore boundary delineation

Fig. 11 shows that the algorithm (right column) produces contours with higher-frequency features than the petrography experts' smoother, continuous boundaries (left column).

This led to underestimating the total pore area, relative to the area identified by experts, for fourteen out of the twenty-five microstructure images as seen in Fig. 11a. There were, however, certain cases where the algorithm detected broader pore outlines compared to the petrography experts and thus a higher overall porosity coming from the large-entrapped pores whose irregular boundaries were exaggerated as seen in Fig. 11b. This could happen due to grey-shadow effects, possibly from secondary light sources, that created ambiguous contrast, causing overestimation of the pore extent.

Petrography experts' conservative annotations avoided these issues, isolating individual pores. These anomalies underscore the system's sensitivity to complex microstructures and lighting, necessitating multi-angle lighting or advanced contrast filters in future iterations.

3.3. Pore location mismatches

In two splits, namely (1,1) and (5,4), our algorithm and the

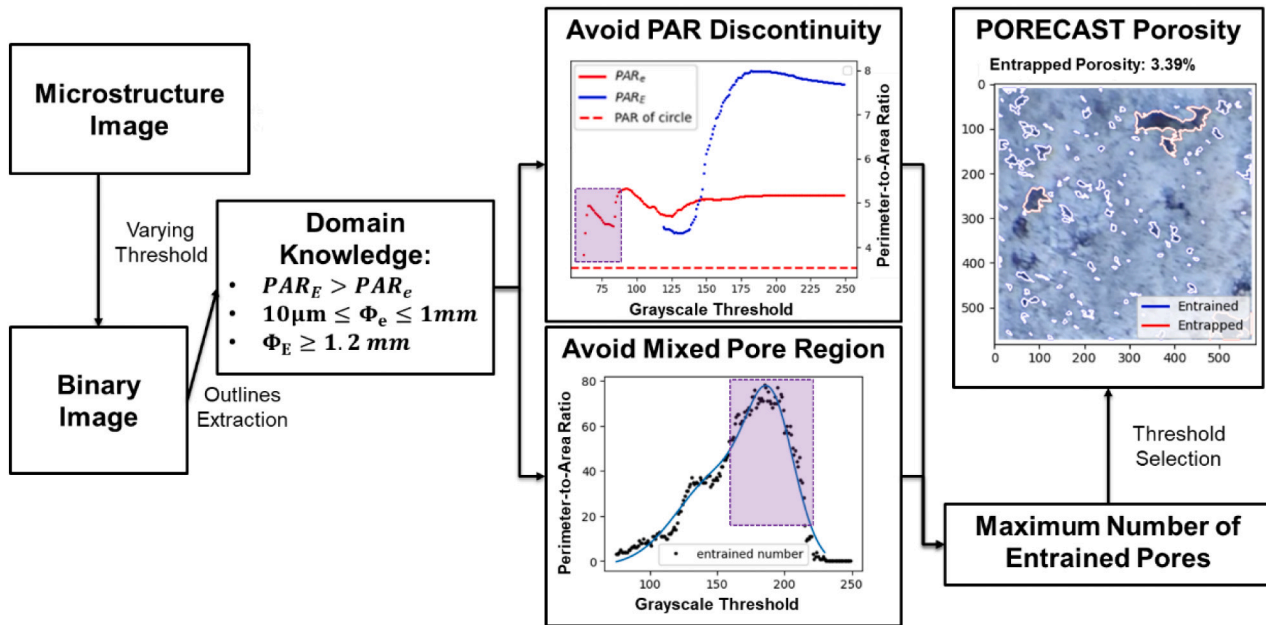


Fig. 8. PORECAST flowchart.

petrography experts identified the largest pores at different locations, yet the overall porosity estimation of the whole area achieved 94.16% and 83.00% accuracy, respectively, compared to experts' porosity estimates. This occurred due to multiple entrapped pores of similar size (>1.25 mm) within these microstructure splits. Experts prioritized prominent pores, while the algorithm detected all pores, sorting them by size. For example, in (1,1) Fig. 12a, one of the petrography experts determined that the actual pore outline, located in the upper-right section of the image, was smaller than the overall dark region. In contrast the second petrography expert annotated the whole dark-grey region as a pore. The algorithm did a similar classification as the second expert while including a smaller pore in the vicinity in the boundary.

For the microstructure (5,4) in Fig. 12b one of the petrography experts determined that no large-entrapped pores exist while the second one identified "loose" light-grey boundaries as three pores. In this case the algorithm had a more conservative identification of the pores with tighter boundaries.

3.4. Optimal threshold extraction

The bimodal distribution of the small-entrained pore count sometimes had overlapping peaks, which complicate an accurate identification of the optimal threshold. Fig. 9 illustrates three instances where the second, larger peak overshadows the smaller first peak, which is critical for detecting small-entrained pores (<1 mm). This overlap, driven by a high signal-to-noise ratio, can lead the double Gaussian regression (Eq. (2)) misidentifying the first peak and require a manual correction. These corrections involve visual assessment of small-entrained pore counts and application of the four rules outlined in Section 2.2 to ensure robust threshold selection. Better regression algorithms and models, based on threshold history, should be incorporated in future improvements of the fully automated algorithm.

For example, in microstructure split (1,1) and (1,4) (Fig. 9a & b), Rule 1 was applied to select the threshold where the PAR of large-entrapped pores (>1.25 mm) becomes continuous and stabilized, ensuring that all large pores are "in-focus" in the binary image. Similarly, in split (1,3) (Fig. 9c) noisy data at high threshold values (~ 150) caused the Gaussian regression to misidentify the first peak. Manual intervention, guided by visual inspection and Rule 3 (excluding the high-threshold mixed region), was used to select a threshold that

accurately captures the overall microstructure. Similarly, in split (1,2) (Fig. 9d) the first distribution is overshadowed from the main second one making it difficult to detect its peak. These cases highlight the importance of integrating domain knowledge and manual oversight to enhance the system's robustness for challenging microstructures.

3.5. Validation summary

The developed monitoring system achieved a 98% porosity prediction accuracy compared to the petrography experts. The 2-pixel lower bound ($10\mu m$) optimized small pore detection, while self-calibration addressed threshold variability. Challenges in boundary delineation and lighting effects highlight areas for refinement, such as integrating convolutional neural networks (CNNs) or multi-spectral imaging to enhance robustness.

Fig. 13b shows the algorithm's results for the individual images compared to the petrography experts' assessment. The boxplots indicate high dispersion of the porosity assessment between experts and the singular value of PORECAST porosity (maroon dot) falls either within that spread or very close for the majority of the cases.

3.6. Porosity characterization

The optimized and validated algorithm enables a detailed characterization of porosity and void geometry in 3D-printed concrete extrudates. This section presents key insights into the size, shape, and distribution of pores, derived from the algorithm's analysis of microstructure images.

Fig. 14a illustrates the size distribution of large-entrapped pores (>1.25 mm), which follows a geometric distribution. Smaller pores are more frequent than larger ones, reflecting typical void patterns on the surfaces of extruded concrete. Fig. 14b plots the circularity of pores using the PAR (Eq. (1)), revealing that most pores are nearly circular (low PAR values), with only a small fraction exhibiting oblong or crack-like shapes (high PAR values). This aligns with concrete domain knowledge, where entrained pores (<1 mm) are typically spherical, and larger entrapped pores may show irregular geometries.

Fig. 14c provides a more comprehensive view of the analysis by integrating combining pore size, circularity (PAR), and frequency in a 3D histogram. This visualization highlights the predominance of small,

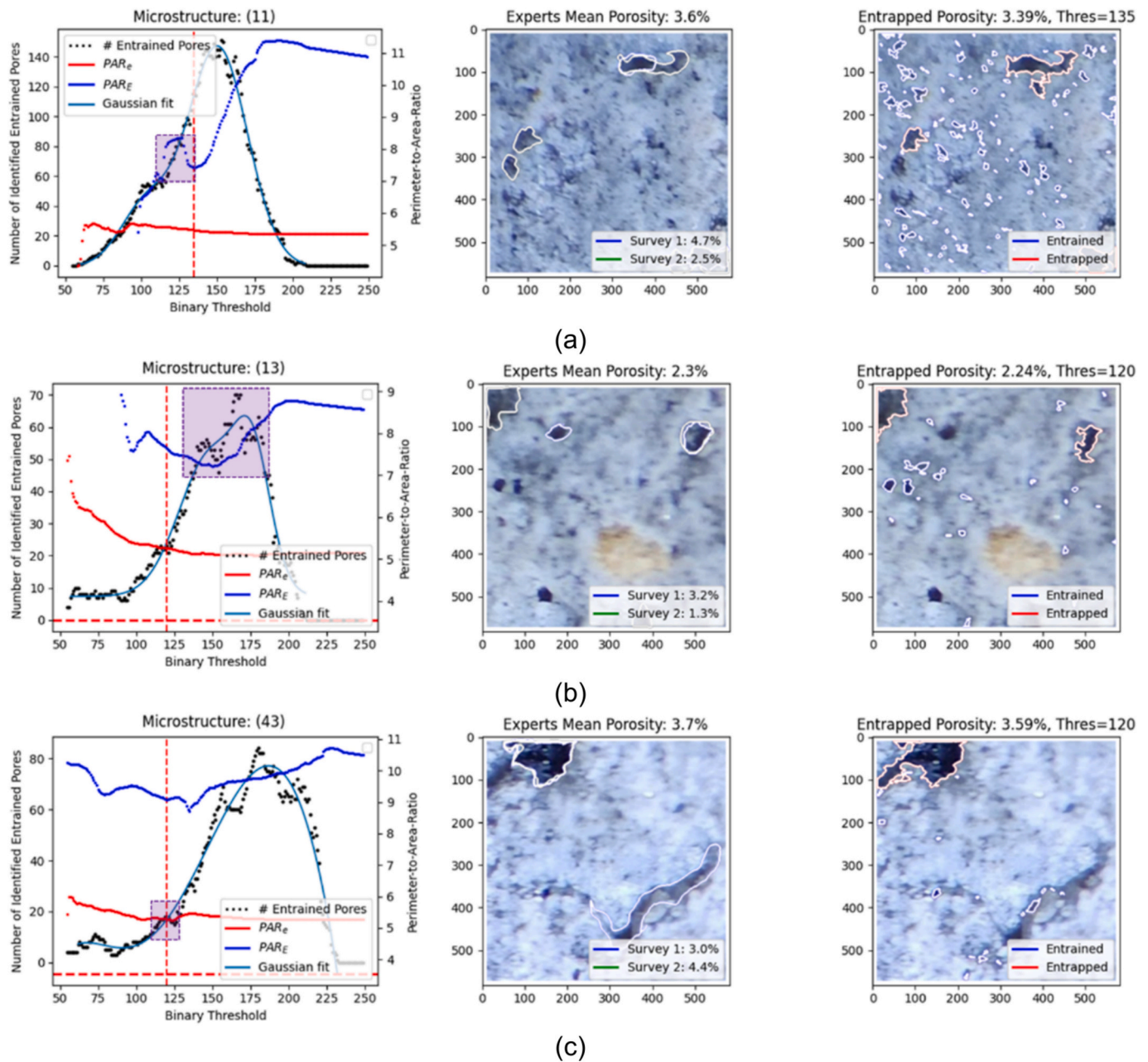


Fig. 9. Instances of bimodal Gaussian distribution challenges in threshold selection: (a) Discontinuous PAR for entrapped voids below 125. (b) Noisy data at 140–180 misleads peak placement. (c) First peak nearly obscured by dominant distribution. Note: x- and y-axes in the microstructure images are pixel counts.

circular pores and the rarity of large, irregular voids, offering valuable insights for optimizing concrete mixture designs and improving structural integrity. These results demonstrate PORECAST's ability to quantify porosity characteristics in real-time, supporting quality control and predictive maintenance in additive construction.

4. Discussion

The development of the PORECAST algorithm addresses a critical void in the current literature regarding the real-time quality assurance of 3D-printed Ultra-High-Performance Concrete (UHPC). While extensive research has successfully quantified the porosity of aerated, heavy-weight, and conventional concretes, these established studies predominantly rely on ex-situ, post-curing methodologies or automated optical traverses on hardened, polished cross-sections. PORECAST is not an incremental optimization of an existing in-process tool; rather, it

establishes a first-of-its-kind operational baseline for this specific material application. While the assessment of internal concrete porosity via surface imaging inherently relies on a two-dimensional representation, this approach aligns directly with the established industry standards for concrete petrography. Traditional methodologies for characterizing a 3D entrained air void system, such as the linear traverse or modified point-count methods outlined in ASTM C457, fundamentally rely on statistical evaluations of 2D planar cross-sections to represent bulk volumetric properties [32]. Recent advancements have introduced sophisticated computational methods capable of mathematically reconstructing or extrapolating 3D void networks from 2D image data, such as the techniques demonstrated by Das et al. (2022) [38] for assessing the air void system of 3D printed concrete. However, integrating complex 2D-to-3D volumetric modeling is highly computationally intensive and falls outside the scope of the current study [39]. This limitation underscores the exact necessity of the PORECAST system: it provides the

Table 1
Methodological framework.

Category	Variable/parameter	Description & operating range
I. Physical experimental parameters (inputs)	Material composition	Ultra-High-Performance Concrete (UHPC) baseline mixture (cast samples).
	Surface preparation	Freshly cut surfaces & top surfaces
	Hardware & illumination	Optical microscopy paired with low-angle, side-mounted illumination to cast shadows in surface depressions.
II. Algorithmic processing variables	Perimeter-to-area ratio (PAR)	Shape descriptor utilized to mathematically distinguish spherical entrained air from irregular entrapped voids.
	Bimodal Gaussian threshold	Self-calibrated to maximize the detection of spherical entrained pores by utilizing the pores bimodal distribution
	Feature filtering range	Size and geometric rules used to prevent misclassification
III. Determined indicators (outputs)	Macroporosity fraction	The calculated percentage of surface area occupied by entrapped voids.
	Void size distribution	The quantitative spread of pore diameters across the analyzed surface domain.
	Morphological characteristics	Circularity and shape metrics of the identified voids to assess their origin
IV. Validation metrics	Baseline ground truth	Expert manual petrographic annotation of the exact surface domain

critical steppingstone to transition the industry away from subjective, laborious manual surveys toward an objective, mathematically

standardized, and automated protocol.

Ultimately, the introduction of this self-calibrating methodology represents a necessary paradigm shift for additive manufacturing in construction. By successfully distinguishing complex void morphologies in real-time—without the need for destructive testing, specimen polishing, or computationally prohibitive hardware—PORECAST elevates UHPC quality control from a reactive, post-process evaluation to a proactive, in-process diagnostic. This capability can unlock the immediate potential for on-the-fly mixture optimization, drastically reducing material waste during extrusion. More importantly, it lays the essential diagnostic groundwork for integrating true, data-driven digital twins into the future of automated construction, ensuring structural integrity from the very first printed layer.

5. Conclusions

This study presented proof-of-concept for PORECAST, a self-calibrating image processing algorithm utilizing concrete domain knowledge, that enables in-process porosity monitoring during 3D concrete printing. The key findings and scientific outcomes of this research are detailed below:

- 1. **Precise Porosity Characterization:** The developed algorithm successfully distinguishes air voids from dark aggregates in 2D surface microstructures, overcoming common misclassification issues in conventional image processing. This enables a highly accurate assessment of pore geometry, size distribution, and circularity.
- 2. **Self-Calibrating Bimodal Thresholding:** The implementation of a Perimeter-to-Area Ratio (PAR) metric within a double Gaussian distribution model allows the system to dynamically self-calibrate. It mathematically isolates small-entrained spherical pores ($< 1\text{ mm}$)

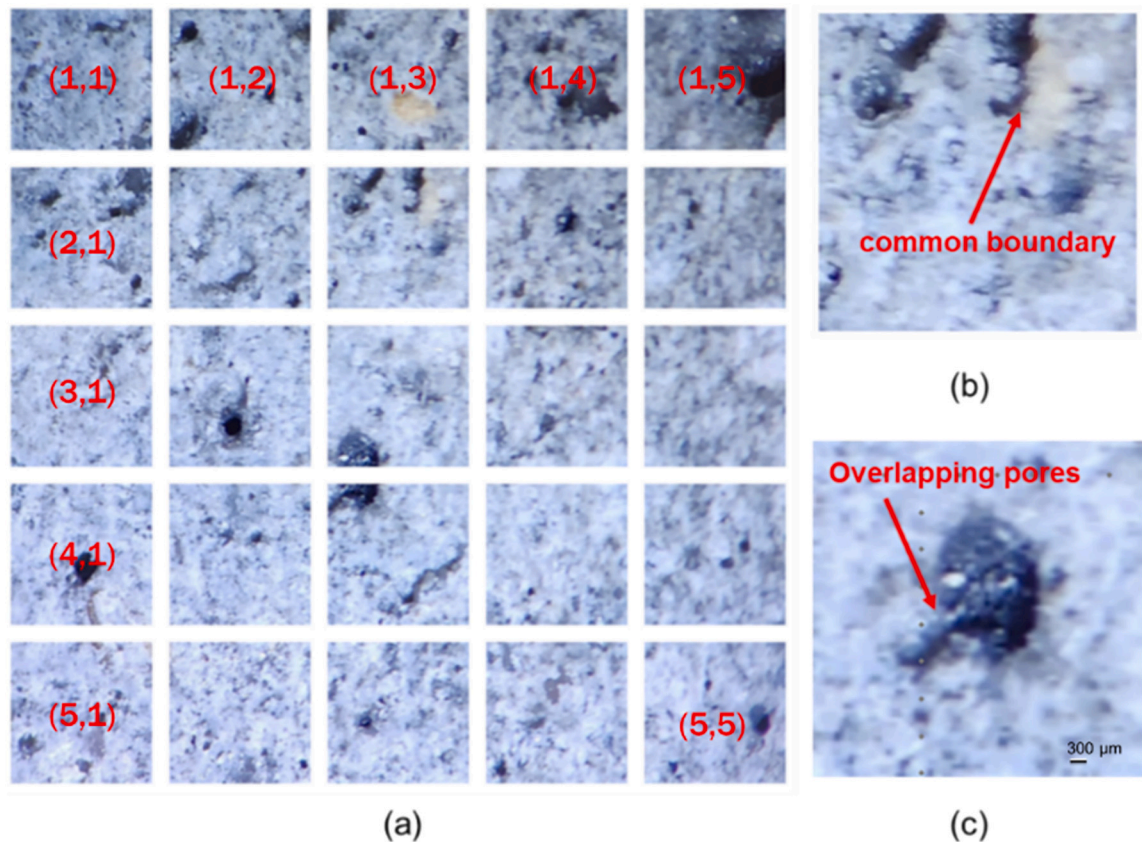


Fig. 10. (a) 5×5 grid of microstructure image. (b) Shared boundary of air void and sand particle. (c) Overlapping air voids.

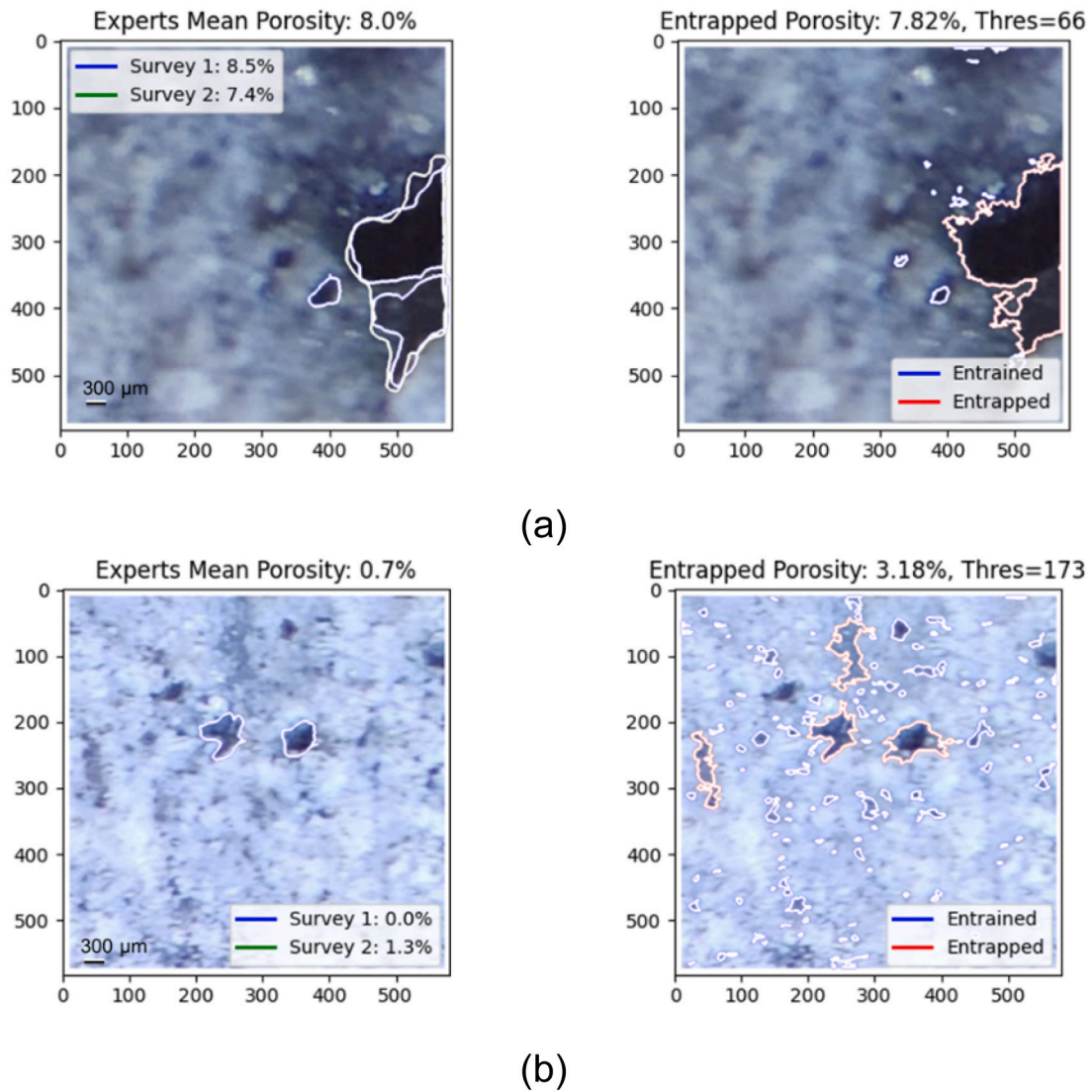


Fig. 11. (a) Instance where PORECAST's pore outlines resulted in tighter boundaries and underestimated the overall porosity area compared to the petrography experts. (b) Instance where the algorithm merged adjacent small pores that shared boundaries, forming single contours with greater area. Note: x- and y-axes in the microstructure images are pixel counts.

from large-entrapped irregular voids, establishing an optimized binarization threshold for feature extraction.

- 3. Validation Against Petrographic Standards:** When validated against expert petrographic annotations, the algorithm's porosity predictions demonstrated high fidelity, proving the scientific viability of utilizing automated 2D optical monitoring as a quantitative diagnostic tool for fresh concrete.

Beyond establishing a robust scientific framework for automated porosity characterization, the deployment of the PORECAST system offers tangible operational value for the 3D concrete printing industry. Translating these fundamental microstructural insights into actionable, real-time data yields the following practical applications:

- 1. Real-Time Quality Assurance:** By extracting immediate porosity data without laborious ex-situ post-processing, the system provides actionable feedback for the on-the-fly optimization of concrete mixture designs, thereby reducing material waste during the extrusion process.
- 2. Predictive Durability and Digital Twins:** Continuous in-process porosity monitoring generates the critical data foundation

necessary to forecast a printed structure's long-term performance (such as freeze-thaw resistance and compressive strength) and facilitates the creation of high-fidelity digital twins for predictive maintenance.

- 3. Scalable Hardware Integration:** The methodology relies on accessible optical hardware (microscopy paired with side-mounted illumination) rather than computationally heavy or expensive alternatives like micro-CT. This allows for cost-effective integration into existing robotic printers (e.g., CyBe Robot Crawler) for rapid deployment in terrestrial, military, and extraterrestrial applications.

Despite its demonstrated efficacy, the current system is at Technology Readiness Level (TRL) 3 (proof-of-concept validation) and carries several limitations. Primarily, the assessment of internal concrete porosity via surface imaging inherently relies on a 2D representation, which requires further computational modeling to extrapolate 3D volumetric void networks. Additionally, the optical capture is sensitive to variations in ambient lighting conditions. Algorithmically, overlapping peaks in the bimodal Gaussian distribution can occasionally complicate the extraction of the optimal threshold, sometimes necessitating manual correction. Finally, this initial validation was restricted to

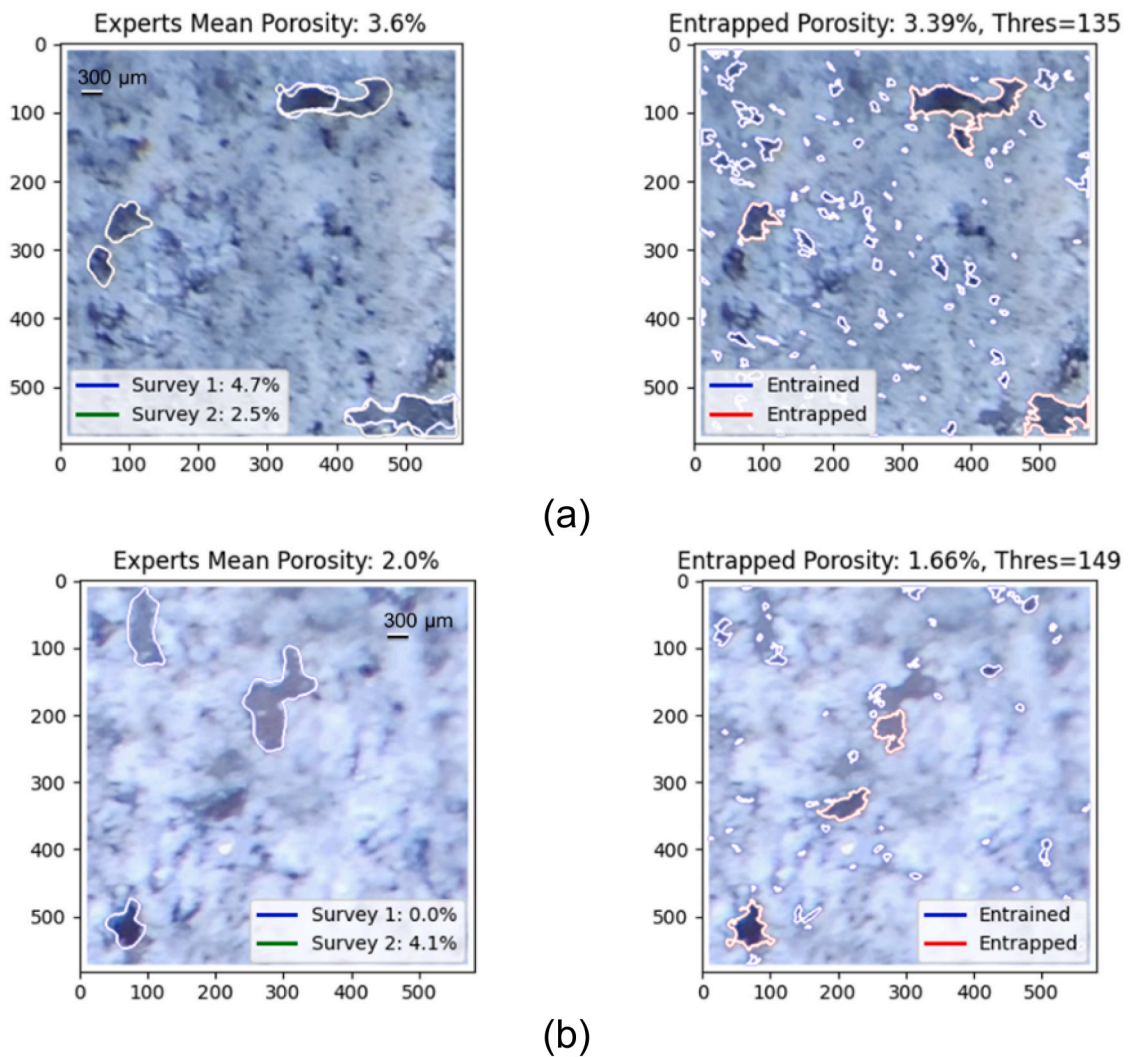


Fig. 12. Mismatches in pore location of the large-entrapped voids between PORECAST and petrography experts. Microstructure split: (a) (1,1) and (b) (5,4). Note: x- and y-axes in the microstructure images are pixel counts.

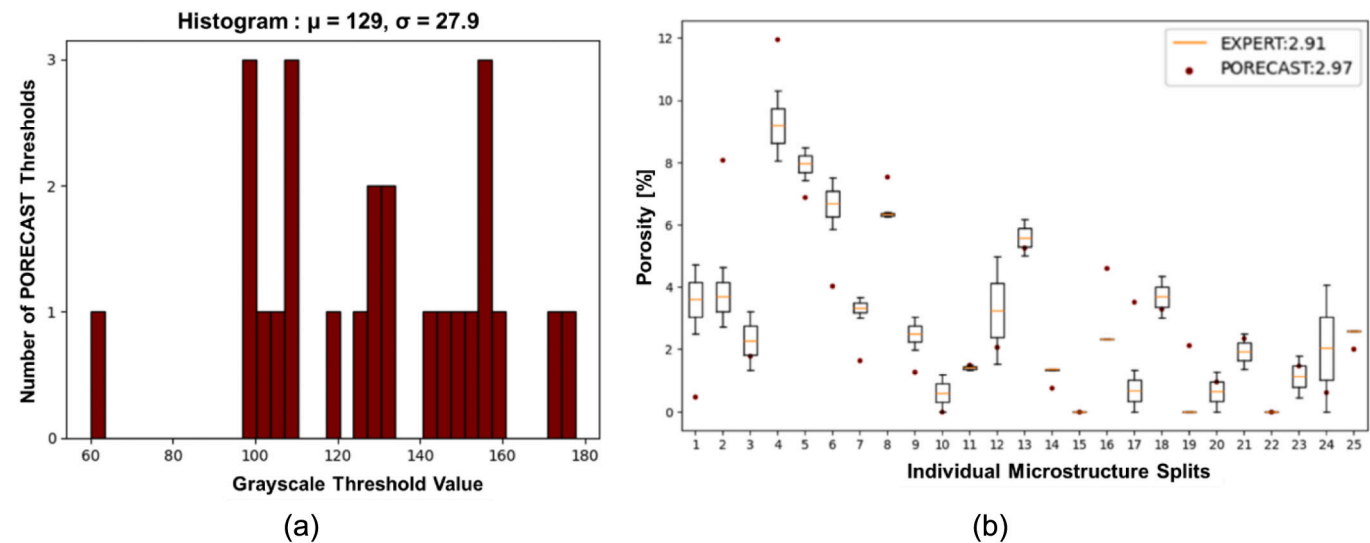


Fig. 13. (a) Histogram of the varying optimal threshold for all processed microstructure images. (b) Comparison study between the PORECAST porosity predictions per split microstructure image and the expert identified porosity.

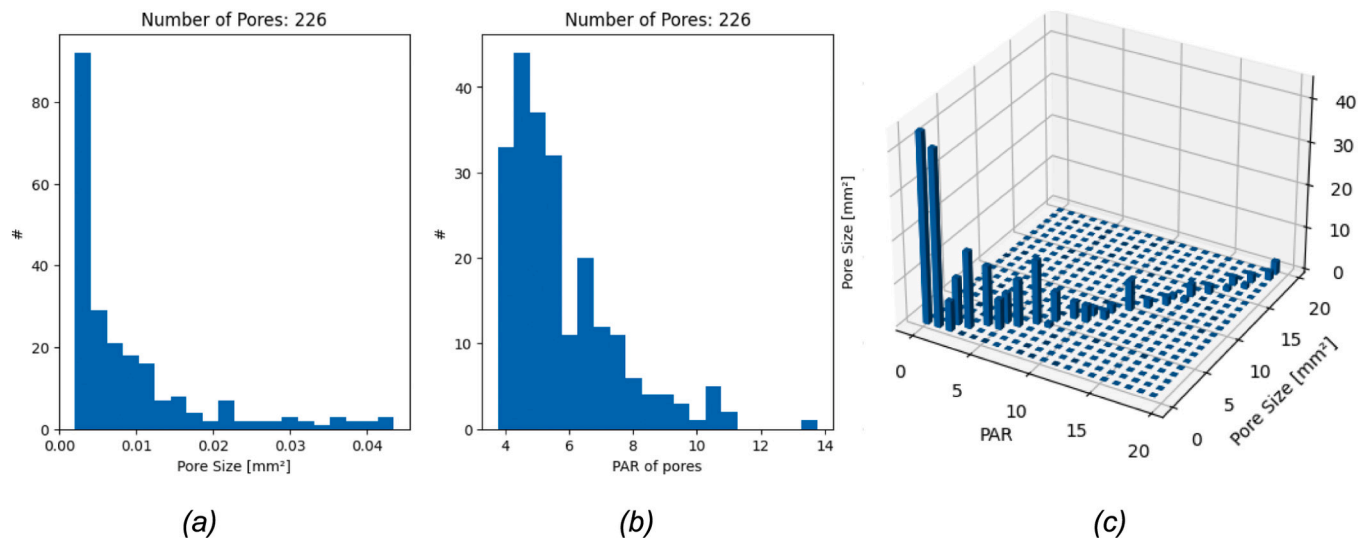


Fig. 14. Porosity key performance indicators (a) number of pores and their size (b) number of pores and their shape (lower PAR means more circular pores) (c) histogram combining pore size, circularity and number of occurrences.

static, cast UHPC samples; therefore, the dynamic physical artifacts introduced by an actively moving 3D printer nozzle remain uncharacterized.

Future work will focus on addressing these limitations by testing the monitoring system on actively extruded filaments in a live 3D printing environment. This will be critical for correlating fresh surface defects directly with final interlayer macro-porosity. Further algorithmic advancements will aim to implement fully automated fitting models to resolve overlapping distributions without manual intervention, alongside the integration of multi-spectral imaging to enhance robustness against variable lighting. Ultimately, expanding the system's validation to encompass a broader range of non-UHPC concrete mixtures and other additive manufacturing materials will help establish this methodology as a versatile industry standard for in-situ monitoring.

CRediT authorship contribution statement

Andreas K. Lianos: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jeffrey W. Bullard:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation. **Dimitris C. Lagoudas:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Satish T.S. Bukkapatnam:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the first author used ChatGPT of OpenAI, Microsoft Bing Copilot and Grok to improve the English and writing style of this paper. After using this tool/service, all the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dr. Satish Bukkapatnam and Andreas Lianos are co-inventors on a pending patent (Application #6756TEES25: "In-Process Porosity

Monitoring System with a Self-Calibrating Computer Vision Algorithm") submitted via the Texas A&M University System. Additionally, Dr. Jeffrey Bullard serves as an editorial board member for Cement and Concrete Research and will be recused from the peer-review process for this manuscript. All other authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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